

JHELUM, PakistanWMO: **415980**Lat: **32.933N**Lon: **73.717E**Elev: **234**StdP: **98.55**Time Zone: **5.00 (E05)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.6	4.9	-1.0	3.6	12.2	0.8	4.1	10.7	3.6	14.4	3.2	15.6	0.0	N/A	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	12.1	42.6	23.5	40.8	23.4	39.2	23.7	29.3	35.2	28.9	34.5	28.4	33.9	1.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.0	24.8	33.0	27.5	24.1	32.7	27.1	23.5	32.3	98.8	34.9	96.6	34.3	94.3	33.9	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.6	3.4	2.7	1.3	45.3	2.1	1.3	-0.2	46.2	-1.5	46.9	-2.7	47.6	-4.2	48.5		
			DB													
			WB													
			1.1	30.6	1.6	0.7	0.0	31.1	-0.9	31.5	-1.8	31.8	-3.0	32.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.9	12.1	15.7	20.6	26.6	30.9	32.7	31.1	30.3	29.1	25.2	18.8	13.6
	DBStd	7.53	2.30	2.66	3.24	2.93	2.74	2.79	2.27	1.92	1.86	2.66	2.23	2.19
	HDD10.0	9	7	1	0	0	0	0	0	0	0	0	0	2
	HDD18.3	458	193	82	16	0	0	0	0	0	0	0	20	147
	CDD10.0	5092	72	159	329	497	649	682	653	628	573	471	265	113
	CDD18.3	2499	1	7	86	247	391	432	394	370	323	213	35	0
	CDH23.3	33315	5	77	795	2854	5715	6787	5743	5087	3902	2024	319	8
CDH26.7	19014	0	6	282	1578	3636	4511	3349	2742	1942	914	54	0	
Wind	WSAvg	0.6	0.4	0.6	0.7	0.8	0.9	1.0	0.8	0.6	0.5	0.3	0.3	0.3
Precipitation	PrecAvg	824	33	49	46	37	28	60	226	221	75	15	10	22
	PrecMax	1239	102	159	170	135	85	191	501	546	252	85	51	120
	PrecMin	453	0	0	0	0	0	4	64	41	0	0	0	0
	PrecStd	185	27	40	42	33	20	45	103	96	64	20	12	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.6	27.5	34.7	39.9	44.0	44.8	41.5	37.4	36.7	35.3	29.8	24.4
		MCWB	14.5	16.8	19.6	21.4	22.5	23.3	27.0	28.6	27.2	24.9	19.0	15.5
	2%	DB	21.3	25.5	31.8	38.0	42.1	43.2	38.9	36.5	35.7	34.1	28.3	22.6
		MCWB	13.1	15.3	19.9	21.0	22.9	23.6	27.7	28.6	26.7	24.2	18.7	14.7
	5%	DB	19.8	23.8	29.8	36.1	40.4	41.7	37.3	35.6	34.8	33.0	26.9	21.2
		MCWB	12.5	14.9	19.0	20.5	22.6	24.0	27.6	28.3	26.5	23.5	18.0	14.1
	10%	DB	18.1	22.0	27.7	34.2	38.7	40.0	35.9	34.7	33.9	31.6	25.3	19.7
		MCWB	11.9	14.2	17.9	20.4	22.3	24.1	27.7	28.0	26.2	22.5	17.3	13.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	18.5	22.8	24.7	26.8	29.0	30.0	29.7	29.2	27.2	21.4	17.0
		MCDB	21.0	25.1	30.6	34.5	37.9	36.9	37.0	35.4	35.0	33.5	27.5	22.3
	2%	WB	14.3	17.0	21.1	23.1	25.3	28.1	29.3	29.3	28.2	25.9	20.3	15.8
		MCDB	19.5	23.4	29.5	34.1	36.7	35.9	35.5	34.8	33.7	31.9	26.3	21.4
	5%	WB	13.4	16.0	20.1	22.1	24.5	27.4	28.8	29.0	27.7	25.0	19.2	14.8
		MCDB	18.2	21.9	27.4	32.8	36.0	35.3	34.6	34.2	33.0	30.5	25.3	20.0
	10%	WB	12.5	14.9	19.0	21.3	23.7	26.7	28.4	28.5	27.1	24.0	18.0	13.7
		MCDB	16.9	20.2	26.0	31.6	35.0	34.6	34.0	33.5	32.2	29.8	23.7	18.7
Mean Daily Temperature Range	5% DB	MDBR	11.7	11.5	12.2	14.0	14.2	12.1	8.2	7.7	9.4	12.6	13.5	12.9
		MCDBR	15.0	14.5	15.2	16.8	16.7	15.0	10.3	8.9	10.4	12.9	15.1	14.9
		MCWBR	8.0	6.4	5.9	5.4	5.1	4.3	3.7	3.6	4.0	5.2	7.3	8.2
	5% WB	MCDBR	12.8	12.1	13.8	15.5	14.8	10.6	8.5	8.4	9.1	11.4	13.3	13.9
		MCWBR	7.4	6.1	6.3	5.9	6.0	4.5	3.7	3.7	4.1	5.2	7.3	8.3
Clear-Sky Solar Irradiance	taub	0.456	0.497	0.506	0.543	0.588	0.679	0.738	0.686	0.591	0.570	0.524	0.470	
	taud	1.939	1.893	1.902	1.858	1.779	1.641	1.586	1.681	1.828	1.769	1.806	1.915	
	Ebn at Noon	719	737	772	763	733	667	624	648	691	659	641	674	
	Edh at Noon	151	175	187	204	222	255	268	240	199	196	172	148	
All-Sky Solar Radiation	RadAvg	2.91	3.63	4.97	6.10	7.01	6.88	5.75	5.36	5.32	4.74	3.57	2.96	
	RadStd	0.32	0.47	0.45	0.34	0.29	0.37	0.45	0.40	0.29	0.23	0.19	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page